EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let 'E' be the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ & 'C' be the circle $x^2+y^2=9$. Let P & Q be the points (1, 2) and (2,1) respectively. Then :

(A) Q lies inside C but outside E (B) Q lies outside both C & E

(C) P lies inside both C & E (D) P lies inside C but outside E.

MEL001

2. The equation, $2x^2+3y^2-8x-18y+35=K$ represents

(A) no locus if $K>0$ (B) an ellipse if $K<0$

(C) a point if $K=0$ (D) a hyperbola if $K>0$

MEL002

3. If $P=(x,y)$, $F_1=(3,0)$, $F_2=(-3,0)$ and $16x^2+25y^2=400$, then PF_1+PF_2 equals

(A) 8 (B) 6 (C) 10 (D) 12

MEL003

4. An ellipse has OB as a semi-minor axis. F and F' are its foci and the angle FBF' is a right angle. Then, the eccentricity of the ellipse is

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

MEL004

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5. The $y-$axis is the directrix of the ellipse with eccentricity $e=1/2$ and the corresponding focus is at (3,0), equation to its auxiliary circle is

(A) $x^2+y^2-8x+12=0$ (B) $x^2+y^2-8x-12=0$

(C) $x^2+y^2-8x+9=0$ (D) $x^2+y^2=4$

MEL005

6. Imagine that you have two thumbtacks placed at two distinct points, A and B. If the ends of a fixed length of string are fastened to the thumbtacks and the string is drawn taut with a pencil, the path traced by the pencil will be a curve called ellipse. The best way to maximise the area surrounded by the curve with a fixed length of string occurs when

I the two points A and B have the maximum distance between them.

II two points A and B coincide.

III A and B are placed vertically.

IV The area is always same regardless of the location of A and B.

(A) I (B) II (C) III (D) IV

MEL006

7. The latus rectum of a conic section is the width of the function through the focus. The positive difference between the length of the latus rectum of $3y=x^2+4x-9$ and $x^2+4y^2-6x+16y=24$ is-

(A) $\frac{1}{2}$ (B) 2 (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

MEL007

For More Join: @IITJEE_Advanced

8. Let $S(5,12)$ and $S'(-12,5)$ are the foci of an ellipse passing through the origin. The eccentricity of ellipse equals -

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{2}{3}$

MEL008

9. A circle has the same centre as an ellipse & passes through the foci F_1 & F_2 of the ellipse, such that the two curves intersect in 4 points. Let 'P' be any one of their point of intersection. If the major axis of the ellipse is 17 & the area of the triangle PF_1F_2 is 30, then the distance between the foci is:

(A) 11 (B) 12 (C)13 (D) none

MEL009

10. $x-2y+4=0$ is a common tangent to $y^2=4x$ & $\frac{x^2}{4}+\frac{y^2}{b^2}=1$. Then the value of b and the other common tangent are given by:

(A) $b=\sqrt{3}\,;x+2y+4=0$ (B) $b=3;x+2y+4=0$

(C) $b=\sqrt{3}\,;x+2y-4=0$ (D) $b=\sqrt{3}\,;x-2y-4=0$

MEL010

11. Consider the particle travelling clockwise on the elliptical path $\frac{x^2}{100}+\frac{y^2}{25}=1$. The particle leaves the orbit at the point (–8, 3) and travels in a straight line tangent to the ellipse. At what point will the particle cross the $y-$axis ?

(A) $\left(0,\frac{25}{3}\right)$ (B) $\left(0,\frac{23}{3}\right)$ (C) $(0,9)$ (D) $\left(0,\frac{26}{3}\right)$

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MEL011

12. The number of values of c such that the straight line $y=4x+c$ touches the curve $\frac{x^2}{4}+y^2=1$ is

(A) 0 (B) 2 (C) 1 (D) ∞

MEL012

13. The normal at a variable point P on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e' such that:

(A) e' is independent of e (B) $e'=1$

(C) $e'=e$ (D) $e'=1/e$

MEL013

14. Suppose x and y are real numbers such that $x^2+9y^2-4x+6y+4=0$ then maximum value of $(4x-9y)$ is

(A) 10 (B) 16 (C) 32 (D) 22

MEL014

15. An ellipse has foci at $F_1(9,20)$ and $F_2(49,55)$ in the xy-plane and is tangent to the x-axis, then the length of its major axis is

(A) 85 (B) 170 (C) 70 (D) 140

MEL015

For More Join: @IITJEE_Advanced

16. The locus of the middle point of chords of an ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ passing through $P(0,5)$ is another ellipse E. The coordinates of the foci of the ellipse E, is

(A) $\left(0,\frac{3}{5}\right)$ and $\left(0,\frac{-3}{5}\right)$ (B) (0, – 4) and (0, 1)

(C) (0, 4) and (0, 1) (D) $\left(0,\frac{11}{2}\right)$ and $\left(0,\frac{-1}{2}\right)$

MEL118

17. A tangent having slope $-\frac{4}{3}$ to the ellipse $\frac{x^2}{18}+\frac{y^2}{32}=1$, intersects the axis of x & y in points A & B respectively. If O is the origin, then area of triangle OAB is.

(A) 12 sq. units (B) 36 sq. units

(C) 24 sq. units (D) 30 sq. units

MEL119

18. Tangents drawn from the point $P(2,3)$ to the circle $x^2 + y^2 - 8x + 6y + 1 = 0$ touch the circle at the points A and B. The circumcircle of the ΔPAB cuts the director circle of ellipse $\frac{(x+5)^2}{9}+\frac{(y-3)^2}{b^2}=1$ orthogonally, then the value of b^2 is

(A) 44 (B) 24 (C) 64 (D) 54

MEL120

19. The equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$ is

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(A) $(x+1)^2 + y^2 = \frac{11}{3}$ (B) $(x-1)^2 + y^2 = \frac{11}{3}$

(C) $(x-1)^2 + y^2 = \frac{11}{3}$ (D) $(x+1)^2 + y^2 = \frac{11}{3}$

MEL121

20. Point 'O' is the centre of the ellipse with major axis AB & minor axis CD. Point F is one focus of the ellipse. If $OF = 6$ & the diameter of the inscribed circle of triangle OCF is 2, then the product $(AB)(CD)$ is

(A) 55 (B) 45 (C) 65 (D) 75

MEL122

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. The equation of the standard ellipse with respect to coordinate axes whose minor axis is equal to the distance between its foci and whose $LR = 10$, will be-

(A) $2x^2 + y^2 = 100$ (B) $x^2 + 2y^2 = 100$

(C) $2x^2 + 3y^2 = 80$ (D) $3x^2 + 2y^2 = 80$

MEL037

2. Consider the ellipse $\frac{x^2}{4}+\frac{y^2}{b^2}=1$ $(b^2 < 4)$ and whose eccentricity is 'e'. If the normal at the end of latus-rectum passes through one extremity of minor axis, then which of the following statement(s) is/are true ?

(A) Equation of director circle of ellipse is $x^2 + y^2 = 10 - 2\sqrt{5}$

(B) Length of latus rectum of the ellipse is equal to $6 - 2\sqrt{5}$

(C) The value of $e^2 + e$ is 1

(D) Area of region between director and auxiliary circle of the ellipse is $2\pi(3-\sqrt{5})$

MEL038

3. Let $A(\alpha)$ and $B(\beta)$ be the extremities of a chord of an ellipse. If the slope of AB is equal to the slope of the tangent at a point $C(\theta)$ on the ellipse, then the value of θ, is

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(A) $\frac{\alpha+\beta}{2}$ (B) $\frac{\alpha-\beta}{2}$

(C) $\frac{\alpha+\beta}{2}+\pi$ (D) $\frac{\alpha-\beta}{2}-\pi$

MEL039

4. Let F_1, F_2 be two foci of the ellipse and PT and PN be the tangent and the normal respectively to the ellipse at point P then

(A) PN bisects $\angle F_1 PF_2$ (B) PT bisects $\angle F_1 PF_2$

(C) PT bisects angle $(180° - \angle F_1 PF_2)$ (D) None of these

MEL040

5. If ℓ_1 be the equation of the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and ellipse $\frac{x^2}{25}+\frac{y^2}{4}=1$ and λ_1 be the length of the intercept of the common tangent between the coordinate axes then

(A) $\lambda_1 = \frac{14}{\sqrt{3}}$ (B) Equation of ℓ_1 is $2x+\sqrt{3}y=4\sqrt{7}$

(C) $\lambda_1 = \frac{4}{\sqrt{3}}$ (D) Equation of ℓ_1 is $x+\sqrt{3}y=4\sqrt{7}$

MEL041

For More Join: @IITJEE_Advanced

6. If latus rectum of an ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ $\{0 < b < 4\}$, subtend angle 2θ at farthest vertex such that $\text{cosec}\theta = \sqrt{5}$, then -

(A) $e=\frac{1}{2}$

(B) no such ellipse exist

(C) $b=2\sqrt{3}$

(D) area of Δ formed by LR and nearest vertex is 6 sq. units

MEL042

7. If $x-2y+k=0$ is a common tangent to $y^2=4x$ & $\frac{x^2}{a^2}+\frac{y^2}{3}=1\left(a>\sqrt{3}\right)$, then the value of a, k and other common tangent are given by -

(A) $a=2$ (B) $a=3$ (C) $x+2y+4=0$ (D) $k=4$

MEL043

8. Equation of the ellipse whose axes are the axes of coordinates and which passes through the point (–3, 1) and has eccentricity $\sqrt{2/5}$ is :-

(A) $3x^2+5y^2-15=0$ (B) $5x^2+3y^2-32=0$

(C) $3x^2+5y^2-32=0$ (D) $5x^2+3y^2-48=0$

MEL044

9. If a and c are positive real number and the ellipse $\frac{x^2}{4c^2}+\frac{y^2}{c^2}=1$ has four distinct points in common with the circle $x^2+y^2=9a^2$, then

(A) $6ac+9a^2-2c^2>0$ (B) $6ac+9a^2-2c^2<0$

(C) $9ac-9a^2-2c^2<0$ (D) $9ac-9a^2-2c^2>0$

MEL045

10. Let the equations of two ellipses be E_1: $\frac{x^2}{3}+\frac{y^2}{2}=1$ and E_2: $\frac{x^2}{16}+\frac{y^2}{b^2}=1$. If the product of their eccentricities is $\frac{1}{2}$, then the length of the minor axis of ellipse E_2 can be :-

(A) 9 (B) 16 (C) 2 (D) 4

MEL046

11. The equation of the common tangents of the parabola $y^2=4x$ and an ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ are -

(A) $x-2y+4=0$ (B) $x+2y+4=0$

(C) $2x-y+1=0$ (D) $2x+y+1=0$

MEL047

12. If length of perpendicular drawn from origin to any normal of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ is ℓ, then ℓ cannot be -

(A) 4 (B) $\frac{5}{2}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

MEL048

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COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 13 to 15

Tangents are drawn from the point $P(3,4)$ to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ touching the ellipse at points A and B.

13. The coordinates of A and B are

(A) (3,0) and (0,2)

(B) $\left(-\frac{8}{5},\frac{2\sqrt{261}}{15}\right)$ and $\left(-\frac{9}{5},\frac{8}{5}\right)$

(C) $\left(-\frac{8}{5},\frac{2\sqrt{161}}{15}\right)$ and (0,2)

(D) (3,0) and $\left(-\frac{9}{5},\frac{8}{5}\right)$

MEL049

14. The orthocenter of the triangle PAB is

(A) $\left(5,\frac{8}{7}\right)$ (B) $\left(\frac{7}{5},\frac{25}{8}\right)$ (C) $\left(\frac{11}{5},\frac{8}{5}\right)$ (D) $\left(\frac{8}{25},\frac{7}{5}\right)$

MEL050

15. The equation of the locus of the point whose distances from the point P and the line AB are equal, is -

(A) $9x^2+y^2-6xy-54x-62y+241=0$

(B) $x^2+9y^2+6xy-54x+62y-241=0$

(C) $9x^2+9y^2-6xy-54x-62y-241=0$

(D) $x^2+y^2-2xy+27x+31y-120=0$

MEL051

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Paragraph for Question No. 16 to 17

Let the two foci of an ellipse be (– 1, 0) and (3, 4) and the foot of perpendicular from the focus (3, 4) upon a tangent to the ellipse be (4, 6).

16. The foot of perpendicular from the focus (–1, 0) upon the same tangent to the ellipse is (α,β) then $5(\alpha+\beta)$ equal to

(A) 46 (B) 47 (C) 48 (D) 49

MEL052

17. The length of semi-minor axis of the ellipse is $\sqrt{\lambda}$ then $\frac{\lambda}{2}$ equal to

(A) 5.50 (B) 6.50 (C) 7.50 (D) 8.50

MEL053

Paragraph for question nos. 18 to 19

Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the parabola $y^2=2x$. They intersect at P and Q in the first and fourth quadrants respectively. Tangents to the ellipse at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

18. The ratio of the areas of the triangles PQS and PQR, is

(A) 1 : 3 (B) 1 : 2 (C) 2 : 3 (D) 3 : 4

MEL123

For More Join: @IITJEE_Advanced

19. The area of quadrilateral $PRQS$, is

(A) $\frac{3\sqrt{15}}{2}$ (B) $\frac{15\sqrt{3}}{2}$ (C) $\frac{5\sqrt{3}}{2}$ (D) $\frac{5\sqrt{15}}{2}$

MEL124

MATRIX MATCH TYPE QUESTION

20.

	Column-I		**Column-II**
(A)	The eccentricity of the ellipse which meets the straight line $2x - 3y = 6$ on the X-axis and the straight line $4x + 5y = 20$ on the y-axis and whose principal axes lie along the coordinate axes, is	(P)	$\frac{1}{2}$
(B)	A bar of length 20 units moves with its ends on two fixed straight lines at right angles. A point P marked on the bar at a distance of 8 units from one end describes a conic whose eccentricity is	(Q)	$\frac{1}{\sqrt{2}}$
(C)	If one extremity of the minor axis of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the foci form an equilateral triangle, then its eccentricity, is	(R)	$\frac{\sqrt{5}}{3}$
(D)	There are exactly two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose distance from the centre of the ellipse are greatest and equal to $\sqrt{\frac{a^2+2b^2}{2}}$. Eccentricity of this ellipse is equal to	(S)	$\frac{\sqrt{7}}{4}$

MEL054

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EXERCISE (S)

1. If $a(x^2 + y^2 + 2y + 1) = (x - 2y + 3)^2$ is an ellipse and $a \in (b, \infty)$, then the value of b is

MEL065

2. An ellipse passes through the points $(-3, 1)$ & $(2, -2)$ and its principal axes are along the coordinate axes in order. The value of $10 \times$ (length of semi-minor axis)^2 is equal to

MEL066

3. Eccentricity of an ellipse in which distance between their foci is 10 and that of focus and corresponding directrix is 15 is e, then 4e is equal to

MEL019

4. If $x, y \in R$, satisfies the equation $\frac{(x-4)^2}{4} + \frac{y^2}{9} = 1$, then the difference between the largest and the smallest value of the expression $\frac{x^2}{4} + \frac{y^2}{9}$ is.

MEL067

5. Rectangle $ABCD$ has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Find the perimeter of the rectangle.

MEL018

6. Equations of the line with equal intercepts on the axes & which touch the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is $lx + my + c = 0$ then value of $25\left(\frac{l^2 + m^2}{c^2}\right)$ is

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MEL016

7. The tangent and normal to the ellipse $x^2 + 4y^2 = 4$ at a point $P(\theta)$ on it meet the major axis in Q and R respectively. If $QR = 2$ and the eccentric angle θ of P is given by $\cos\theta = \pm\left(\frac{a}{b}\right)$, where a & b are relative prime, then value of $\frac{2b}{a}$ is equal to

MEL020

8. A circle intersects an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ precisely at three points A, B, C as shown in the figure. AB is a diameter of the circle and is perpendicular to the major axis of the ellipse. If the eccentricity of the ellipse is $4/5$, length of the diameter AB is λa, then find λ.

MEL068

9. Let P_i and P_i' be the feet of the perpendiculars drawn from foci S, S' on a tangent T_i to an ellipse whose length of semi-major axis is 20. If $\sum_{i=1}^{10}(SP_i)(S'P_1') = 2560$, then the value of $100e$ is (where e eccentricity)

MEL017

10. S_1 and S_2 are the foci of an ellipse of major axis of length 10 units, and P is any point on the ellipse such that the perimeter of triangle PS_1S_2 is 15. Then the eccentricity of the ellipse is

MEL069

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. The area (in sq. units) of the quadrilateral formed by the tangents at the end points of the latera recta to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ is: **[JEE (Main) 2015]**

(1) $\frac{27}{2}$ (2) 27 (3) $\frac{27}{4}$ (4) 18

MEL021

2. The eccentricity of an ellipse whose centre is at the origin is $\frac{1}{2}$. If one of its directices is $x=-4$, then the equation of the normal to it at $\left(1,\frac{3}{2}\right)$ is:- **[JEE (Main) 2017]**

(1) $x+2y=4$ (2) $2y-x=2$ (3) $4x-2y=1$ (4) $4x+2y=7$

MEL022

3. If tangents are drawn to the ellipse $x^2+2y^2=2$ at all points on the ellipse other than its four vertices then the mid points of the tangents intercepted between the coordinate axes lie on the curve: **[JEE (Main) 2019]**

(1) $\frac{x^2}{2}+\frac{y^2}{4}=1$ (2) $\frac{x^2}{4}+\frac{y^2}{2}=1$ (3) $\frac{1}{2x^2}+\frac{1}{4y^2}=1$ (4) $\frac{1}{4x^2}+\frac{1}{2y^2}=1$

MEL023

4. Let S and S' be the foci of the ellipse and B be any one of the extremities of its minor axis. If $\Delta S'BS$ is a right angled triangle with right angle at B and area $(\Delta S'BS)$ = 8 sq. units, then the length of a latus rectum of the ellipse is : **[JEE (Main) 2019]**

(1) $2\sqrt{2}$ (2) 2 (3) 4 (4) $4\sqrt{2}$

MEL024

5. If the line $x-2y=12$ is tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at the point $\left(3,\frac{-9}{2}\right)$, then the length of the latus rectum of the ellipse is : **[JEE (Main) 2019]**

(1) 9 (2) $8\sqrt{3}$ (3) $12\sqrt{2}$ (4) 5

MEL025

6. If the normal to the ellipse $3x^2+4y^2=12$ at a point P on it is parallel to the line, $2x+y=4$ and the tangent to the ellipse at P passes through $Q(4,4)$ then PQ is equal to: **[JEE (Main) 2019]**

(1) $\frac{\sqrt{221}}{2}$ (2) $\frac{\sqrt{157}}{2}$ (3) $\frac{\sqrt{61}}{2}$ (4) $\frac{5\sqrt{5}}{2}$

MEL026

7. If $3x+4y=12\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{9}=1$ for some $a\in R$, then the distance between the foci of the ellipse is : **[JEE (Main) 2020]**

(1) 4 (2) $2\sqrt{7}$ (3) $2\sqrt{5}$ (4) $2\sqrt{2}$

MEL027

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8. If the distance between the foci of an ellipse is 6 and the distance between its directrices is 12, then the length of its latus rectum is: **[JEE (Main) 2020]**

(1) $\sqrt{3}$ (2) $2\sqrt{3}$ (3) $3\sqrt{2}$ (4) $\frac{3}{\sqrt{2}}$

MEL028

9. Let the line $y = mx$ and the ellipse $2x^2 + y^2 = 1$ intersect at a point P in the first quadrant. If the normal to this ellipse at P meets the co-ordinate axes at $\left(-\frac{1}{3\sqrt{2}}, 0\right)$ and $(0,\beta)$, then β is equal to **[JEE (Main) 2020]**

(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{2\sqrt{2}}{3}$ (3) $\frac{2}{3}$ (4) $\frac{\sqrt{2}}{3}$

MEL029

10. The length of the minor axis (along y-axis) of an ellipse in the standard form is $\frac{4}{\sqrt{3}}$. If this ellipse touches the line, $x + 6y = 8$; then its eccentricity is: **[JEE (Main) 2020]**

(1) $\sqrt{\frac{5}{6}}$ (2) $\frac{1}{2}\sqrt{\frac{11}{3}}$ (3) $\frac{1}{3}\sqrt{\frac{11}{3}}$ (4) $\frac{1}{2}\sqrt{\frac{5}{3}}$

MEL030

11. If the point of intersections of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the circle $x^2 + y^2 = 4b, b > 4$ lie on the curve $y^2 = 3x^2$, then b is equal to: **[JEE (Main) 2021]**

(1) 12 (2) 5 (3) 6 (4) 10

MEL031

12. Let L be a tangent line to the parabola $y^2 = 4x - 20$ at (6, 2). If L is also a tangent to the ellipse $\frac{x^2}{2}+\frac{y^2}{b}=1$, then the value of b is equal to: **[JEE (Main) 2021]**

(1) 11 (2) 14 (3) 16 (4) 20

MEL032

13. If m is the slope of a common tangent to the curves $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2 + y^2 = 12$, then $12m^2$ is equal to: **[JEE (Main) 2022]**

(1) 6 (2) 9 (3) 10 (4) 12

MEL033

14. The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with eccentricity: **[JEE (Main) 2022]**

(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{2\sqrt{2}}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{2}$

MEL034

15. Let the ellipse $E: x^2 + 9y^2 = 9$ intersect the positive x-and y-axes at the points A and B respectively. Let the major axis of E be a diameter of the circle C. Let the line passing through A and B meet the circle C at the point P. If the area of the triangle which vertices A, P and the origin O is $\frac{m}{n}$, where m and n are coprime, then $m - n$ is equal to **[JEE (Main) 2023]**

(1) 18 (2) 16 (3) 17 (4) 15

MEL035

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16. Consider ellipses $E_k : kx^2 + k^2y^2 = 1, k = 1, 2, \ldots, 20$. Let C_k be the circle which touches the four chords joining the end points (one on minor axis and another on major axis) of the ellipse E_k, If r_k is the radius of the circle C_k, then the value of $\sum_{k=1}^{20} \frac{1}{r_k^2}$ is **[JEE (Main) 2023]**

(1) 3080 (2) 3210 (3) 3320 (4) 2870

MEL036

17. Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = fog(10)$, $b = gof(3)$. If e and l denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then $8e^2 + l^2$ is equal to.

[JEE (Main) 2024]

(1) 16 (2) 8 (3) 6 (4) 12

MEL125

18. Let P be a point on the ellipse $\frac{x^2}{2} + \frac{y^2}{4} = 1$. Let the line passing through P and parallel to y-axis meet the circle $x^2 + y^2 = 9$ at point Q such that P and Q are on the same side of the x-axis. Then, the eccentricity of the locus of the point R on PQ such that $PR : RQ = 4 : 3$ as P moves on the ellipse, is :

[JEE (Main) 2024]

(1) $\frac{11}{19}$ (2) $\frac{13}{21}$ (3) $\frac{\sqrt{139}}{23}$ (4) $\frac{\sqrt{13}}{7}$

MEL126

19. Let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$ be an ellipse, whose eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is $\sqrt{14}$. Then the square of the eccentricity of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is: **[JEE (Main) 2024]**

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(1) 3 (2) 7/2 (3) 3/2 (4) 5/2

MEL127

20. The length of the chord of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, whose mid point is $\left(1, \frac{2}{5}\right)$, is equal to:

[JEE (Main) 2024]

(1) $\frac{\sqrt{1691}}{5}$ (2) $\frac{\sqrt{2009}}{5}$ (3) $\frac{\sqrt{1741}}{5}$ (4) $\frac{\sqrt{1541}}{5}$

MEL128

21. If the points of intersection of two distinct conics $x^2 + y^2 = 4b$ and $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ lie on the curve $y^2 = 3x^2$, then $3\sqrt{3}$ times the area of the rectangle formed by the intersection points is __

[JEE (Main) 2024]

MEL129

22. Let $A(\alpha, 0)$ and $B(0, \beta)$ be the points on the line $5x + 7y = 50$. Let the point P divide the line segment AB internally in the ratio 7 : 3. Let $3x - 25 = 0$ be a directrix of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the corresponding focus be S. If from S, the perpendicular on the x-axis passes through P, then the length of the latus rectum of E is equal to **[JEE (Main) 2024]**

(1) $\frac{25}{3}$ (2) $\frac{32}{9}$ (3) $\frac{25}{9}$ (4) $\frac{32}{5}$

MEL130

For More Join: @IITJEE_Advanced

23. If the length of the minor axis of ellipse is equal to half of the distance between the foci, then the eccentricity of the ellipse is: **[JEE (Main) 2024]**

(1) $\frac{\sqrt{5}}{3}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{3}}$ (4) $\frac{2}{\sqrt{5}}$

MEL131

24. Let the foci and length of the latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,\ a>b$ be $(\pm 5,0)$ and $\sqrt{50}$, respectively. Then, the square of the eccentricity of the hyperbola $\frac{x^2}{b^2}-\frac{y^2}{a^2b^2}=1$ equals

[JEE (Main) 2024]

MEL132

25. Let the line $2x+3y-k=0, k>0$, intersect the x-axis and y-axis at the points A and B, respectively. If the equation of the circle having the line segment AB as a diameter is $x^2+y^2-3x-2y=0$ and the length of the latus rectum of the ellipse $x^2+9y^2=k^2$ is $\frac{m}{n}$, where m and n are coprime, then $2m+n$ is equal to **[JEE (Main) 2024]**

(1) 10 (2) 11 (3) 13 (4) 12

MEL133

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EXERCISE - JEE (Advanced) PYQ

1. Suppose that the foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ are $(f_1,0)$ and $(f_2,0)$ where $f_1>0$ and $f_2<0$. Let P_1 and P_2 be two parabolas with a common vertex at (0,0) and with foci at $(f_1,0)$ and $(2f_2,0)$, respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2,0)$ and T_2 be a tangent to P_2 which passes through $(f_1,0)$. If m_1 is the slope of T_1 and m_2 is the slope of T_2, then the value of $\left(\frac{1}{m_1^2}+m_2^2\right)$ is

[JEE (Advanced) 2015]

MEL055

2. Let E_1 and E_2 be two ellipses whose centers are at the origin. The major axes of E_1 and E_2 lie along the x −axis and the y −axis, respectively. Let S be the circle $x^2+(y-1)^2=2$. The straight line $x+y=3$ touches the curves S, E_1 and E_2 at P, Q and R, respectively. Suppose that $PQ=PR=\frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2, respectively, then the correct expression(s) is(are)

[JEE (Advanced) 2015]

(A) $e_1^2+e_2^2=\frac{43}{40}$ (B) $e_1e_2=\frac{\sqrt{7}}{2\sqrt{10}}$

(C) $\left|e_1^2-e_2^2\right|=\frac{5}{8}$ (D) $e_1e_2=\frac{\sqrt{3}}{4}$

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MEL056

Paragraph For Question No. 3 to 4

Let $F_1(x_1,0)$ and $F_2(x_2,0)$ for $x_1<0$ and $x_2>0$, be the foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{8}=1$. Suppose a parabola having vertex at the origin and focus at F_2 intersects the ellipse at point M in the first quadrant and at point N in the fourth quadrant.

3. The orthocentre of the triangle F_1MN is- **[JEE (Advanced) 2016]**

(A) $\left(-\frac{9}{10},0\right)$ (B) $\left(\frac{2}{3},0\right)$

(C) $\left(\frac{9}{10},0\right)$ (D) $\left(\frac{2}{3},\sqrt{6}\right)$

MEL057

4. If the tangents to the ellipse at M and N meet at R and the normal to the parabola at M meets the x-axis at Q, then the ratio of area of the triangle MQR to area of the quadrilateral MF_1NF_2 is -

[JEE (Advanced) 2016]

(A) 3 : 4 (B) 4 : 5

(C) 5 : 8 (D) 2 : 3

MEL058

For More Join: @IITJEE_Advanced

5. Consider two straight lines, each of which is tangent to both the circle $x^2+y^2=\frac{1}{2}$ and the parabola $y^2=4x$. Let these lines intersect at the point Q. Consider the ellipse whose center is at the origin $O(0,0)$ and whose semi-major axis is OQ. If the length of the minor axis of this ellipse is $\sqrt{2}$, then the which of the following statement(s) is (are) TRUE ? **[JEE (Advanced) 2018]**

(A) For the ellipse, the eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is 1

(B) For the ellipse, the eccentricity is $\frac{1}{2}$ and the length of the latus rectum is $\frac{1}{2}$

(C) The area of the region bounded by the ellipse between the lines $x=\frac{1}{\sqrt{2}}$ and $x=1$ is

$\frac{1}{4\sqrt{2}}(\pi-2)$

(D) The area of the region bounded by the ellipse between the lines $x=\frac{1}{\sqrt{2}}$ and $x=1$ is $\frac{1}{16}(\pi-2)$

MEL059

6. Define the collections $\{E_1,E_2,E_3,\ldots.\}$ of ellipses and $\{R_1,R_2,R_3,\ldots.\}$ of rectangles as follows :

$E_1:\frac{x^2}{9}+\frac{y^2}{4}=1$;

R_1 : rectangle of largest area, with sides parallel to the axes, inscribed in E_1 ;

E_n : ellipse $\frac{x^2}{a_n^2}+\frac{y^2}{b_n^2}=1$ of largest area inscribed in $R_{n-1}, n>1$;

R_n : rectangle of largest area, with sides parallel to the axes, inscribed in $E_n, n>1$.

Then which of the following options is/are correct ? **[JEE (Advanced) 2019]**

(A) The eccentricities of E_{18} and E_{19} are NOT equal

(B) The distance of a focus from the centre in E_9 is $\frac{\sqrt{5}}{32}$

(C) The length of latus rectum of E_9 is $\frac{1}{6}$

(D) $\sum_{n=1}^{N}(\text{area of } R_n)<24$, for each positive integer N

MEL060

7. Let a, b and λ be positive real numbers. Suppose P is an end point of the latus rectum of the parabola $y^2=4\lambda x$, and suppose the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through the point P. If the tangents to the parabola and the ellipse at the point P are perpendicular to each other, then the eccentricity of the ellipse is **[JEE (Advanced) 2020]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$

MEL061

For More Join: @IITJEE_Advanced

8. Let E be the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$. For any three distinct points P, Q and Q' on E, let $M\ (P, Q)$ be the mid-point of the line segment joining P and Q, and $M(P, Q')$ be the mid-point of the line segment joining P and Q'. Then the maximum possible value of the distance between $M(P, Q)$ and $M(P, Q')$, as P, Q and Q' vary on E, is ___. **[JEE (Advanced) 2021]**

MEL062

9. Consider the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$. Let $H(\alpha, 0)$, $0 < \alpha < 2$, be a point. A straight line drawn through H parallel to the y −axis crosses the ellipse and its auxiliary circle at points E and F respectively, in the first quadrant. The tangent to the ellipse at the point E intersects the positive x −axis at a point G. Suppose the straight line joining F and the origin makes an angle ϕ with the positive x −axis.

	List-I		**List-II**
(I)	If $\phi=\frac{\pi}{4}$, then the area of the triangle FGH is	(P)	$\frac{\left(\sqrt{3}-1\right)^4}{8}$
(II)	If $\phi=\frac{\pi}{3}$, then the area of the triangle FGH is	(Q)	1
(III)	If $\phi=\frac{\pi}{6}$, then the area of the triangle FGH is	(R)	$\frac{3}{4}$
(IV)	If $\phi=\frac{\pi}{12}$, then the area of the triangle FGH is	(S)	$\frac{1}{2\sqrt{3}}$
		(T)	$\frac{3\sqrt{3}}{2}$

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The correct option is : **[JEE (Advanced) 2022]**

(A) (I) → (R); (II) → (S); (III) → (Q); (IV) → (P)

(B) (I) → (R); (II) → (T); (III) → (S); (IV) → (P)

(C) (I) → (Q); (II) → (T); (III) → (S); (IV) → (P)

(D) (I) → (Q); (II) → (S); (III) → (Q); (IV) → (P)

MEL063

10. Let T_1 and T_2 be two distinct common tangents to the ellipse E : $\frac{x^2}{6}+\frac{y^2}{3}=1$ and the parabola $P: y^2 = 12x$. Suppose that the tangent T_1 touches P and E at the point A_1 and A_2, respectively and the tangent T_2 touches P and E at the points A_4 and A_3, respectively. Then which of the following statements is(are) true? **[JEE (Advanced) 2023]**

(A) The area of the quadrilateral $A_1A_2A_3A_4$ is 35 square units

(B) The area of the quadrilateral $A_1A_2A_3A_4$ is 36 square units

(C) The tangents T_1 and T_2 meet the x-axis at the point (–3, 0)

(D) The tangents T_1 and T_2 meet the x-axis at the point (–6, 0)

MEL064

For More Join: @IITJEE_Advanced

11. Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$. Let $S(p,q)$ be a point in the first quadrant such that $\frac{p^2}{9}+\frac{q^2}{4}>1$. Two tangents are drawn from S to the ellipse, of which one meets the ellipse at one end point of the minor axis and the other meets the ellipse at a point T in the fourth quadrant. Let R be the vertex of the ellipse with positive x-coordinate and O be the center of the ellipse. If the area of the triangle ΔORT is $\frac{3}{2}$, then which of the following options is correct ? **[JEE (Advanced) 2024]**

(A) $q=2,\ p=3\sqrt{3}$

(B) $q=2,\ p=4\sqrt{3}$

(C) $q=1,\ p=5\sqrt{3}$

(D) $q=1,\ p=6\sqrt{3}$

MEL134

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	C	C	A	B	A	C	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	C	B	A	C	C	D	B	C

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,D	A,C	A,C	A,B	A,C,D	A,C,D	C,D	A,B	B,D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	A,B	A,B	D	C	A	A	D	C	B	
Que.	20									
Ans.	A→S; B→R; C→P; D→Q									

EXERCISE (S)

1. 5.00 **2.** 64.00 **3.** 2 **4.** 8.00 **5.** 80
6. 2 **7.** 3 **8.** 1.05 or 1.06 **9.** 60 **10.** 0.50

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	3	1	4	2	3	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	2	3	3	1	2	4	3	1
Que.	21	22	23	24	25					
Ans.	432	4	4	51	2					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10	11
Ans.	4	A,B	A	C	A,C	C,D	A	4	C	A,C	A

For More Join: @IITJEE_Advanced